

PRESENTATION SUMMARY: WIND PORTFOLIO PERFORMANCE OPTIMIZATION

1. BUSINESS CONTEXT & STRATEGIC PROBLEM (CRISP-DM STEP 1)

- Asset Base: 1 Gigawatt (1,000 MW) wind portfolio generating ~3.33 TWh/year at a 38% baseline capacity factor.
- The Business Problem: Annual energy generation gap of 5% to 8% fleet-wide. ~14% of operational hours exhibit a power deficit >10% compared to theoretical limits, resulting in multi-million dollar annual revenue losses.
- Operational Constraint: Zero-CapEx mandate. Optimization must be achieved strictly via software, control re-tuning, and targeted O&M dispatching with no new hardware.

2. BUSINESS OBJECTIVES & QUANTIFIED SUCCESS CRITERIA

- Primary Objective: Reclaim 2% to 3% of Annual Energy Production (AEP) by isolating and correcting controllable technical faults.
- Recovery Target: 66.6 to 99.9 GWh/year of recovered energy.
- Technical Benchmark: Baseline regression models must achieve high power prediction accuracy with $R^2 \geq 0.90$ without data leakage.
- Operational Criteria: Eliminate false alarms and rank maintenance dispatches based on true financial revenue impact.

3. PROBLEM FRAMING: QUESTION BANK & LOSS ATTRIBUTION

- Loss Breakdown:
 - ~60% Non-Controllable Losses: Weather extremes (35%) & grid curtailments (25%).
 - ~40% Addressable/Controllable Faults (PRIMARY TARGET): Yaw misalignment (18%), pitch angle drift (14%), and early thermal/component wear (8%).
- Problem Formulation (Core Targets):
 1. What should the turbine produce right now? (Target: Expected_Power_kW)
 2. Is there an addressable deficit? (Target: Underperformance_Flag >10% loss during eligible, non-curtailed operating windows)
 3. What is causing it and what is the financial ROI to fix it? (SHAP attribution & Lost_MWh $\times$ Price ranking)

4. ANALYTICAL UNIT & DATA PIPELINE STRATEGY

- Unit of Analysis: 1 Physical Wind Turbine observed at a 10-Minute SCADA Timestamp.
- 5-Stage Machine Learning Pipeline:
 1. Expected Power Baseline: Time-aware XGBoost/LightGBM model predicting baseline power curve.
 2. Deficit Flagging: Automated filtering for eligible, uncurtailed operating windows.
 3. Loss Attribution: Disambiguates non-controllable environmental factors from technical faults.

4. Early Anomaly Detection: Multi-sensor health tracking (vibration, thermal drift) 14-30 days prior to breakdown.
5. Monetized Prioritization: Ranks underperforming turbines by revenue impact to automate maintenance dispatch.

5. DEPLOYMENT ROADMAP & EXIT GATE

Rolling out the project in four simple phases over 7+ months:

- Months 1–2 (Past Data Test): Train models on historical SCADA data, hitting a strict 90% accuracy target ($R^2 \geq 0.90$).
- Months 3–4 (200 MW Pilot): Test in shadow mode across 2 wind farms, sending technicians to physically verify flagged yaw/pitch issues.
- Months 5–6 (System Integration): Stream live 10-minute SCADA data to directly automate maintenance work orders in CMMS.
- Months 7+ (Full 1 GW Rollout): Scale across the entire 1,000 MW fleet with continuous automated tracking.

STAGE 1 SIGN-OFF:

Stage 1 is complete. With sign-off, we move to Stage 2 (Data Understanding) to unlock 66.6 to 99.9 GWh/year in lost energy with \$0 capital expenditure.